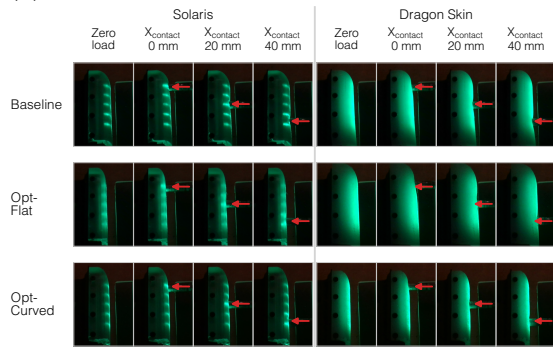
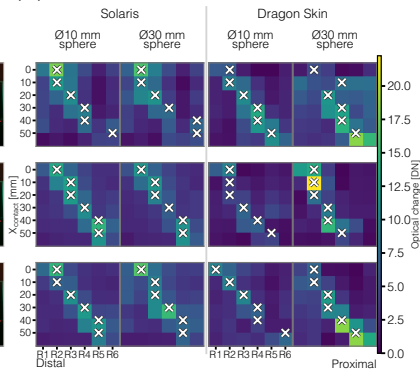


(a) Optical signatures across contact locations



(b) Optical signal change, 2-15 N



(c) Contact-location decoding

